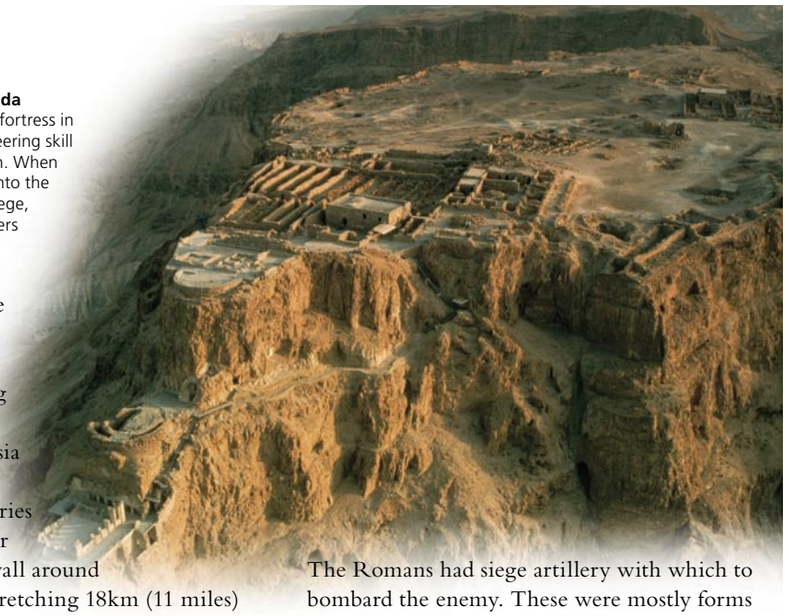
decisive campaign five years later they built a monumental stone-and-wood arched bridge, accessed by an approach road cut into the cliffs lining the river. It has been suggested that the legionaries should be regarded more as combat engineers than as straightforward infantry.

The need to use tools as much as weapons was also much in evidence during siege operations, which formed such an important part of the warfare of the age. Taking a fortified position that was stoutly defended was a challenge to any attacking force, but the Roman legions achieved the feat time and again through a combination of engineering skills and indomitable fighting spirit.

SIEGE TACTICS AND WEAPONS

Sieges were normally long, drawn-out affairs, because an assault on a fortress or fortified town was a desperate business to attempt, even once the walls were breached. Fighting your way into a stronghold was a last resort, and defenders who made it necessary by refusing to surrender could expect no mercy. Victorious legionaries, who had seen colleagues killed or wounded by missiles as they advanced on the fortifications and in the confused close-quarters fighting that followed, indulged in an orgy of slaughter, rape, and pillage. This was the legionary's reward for the hardships and dangers of the siege and his revenge on those who had caused him so much trouble. It was also deliberate Roman policy, designed to deter others from attempting to defy Roman power.



The stronghold of Masada
The taking of the cliff-top fortress in 73 CE showed great engineering skill and ruthless determination. When the Romans finally broke into the fortress after two years' siege, the few remaining defenders had committed suicide.

The scale of the siege works undertaken was often highly impressive. Besieging the Gallic army of Vercingetorix at Alesia in central Gaul in 52 BCE, Caesar's legionaries constructed a circular ditch-and-rampart wall around the Gauls' hill fort stretching 18km (11 miles) and incorporating 23 forts and over 100 wooden towers. Having completed this massive work, they then built an even longer fortified wall facing outward, to defend themselves against a Gallic army arriving to relieve Vercingetorix.

At the siege of the mountain fortress of Masada in Palestine, held by a defiant band of Jewish rebels in 73 CE, the Tenth Legion built a ramp 600m (nearly 2,000ft) long reaching from the foot to the top of the 200-m (650-ft) high sheer rock on which the fortress perched. Constructed under constant fire, this ramp allowed them to haul a giant battering ram up to the fortress and hammer a breach in its walls.

The Romans had siege artillery with which to bombard the enemy. These were mostly forms of *ballista*, a torsion machine superficially resembling a large crossbow. In the late empire period *ballistae* were supplemented by the *onager*, a one-armed catapult. None of these siege engines, however, packed sufficient punch to breach city walls or other solid fortifications. They were usually fired from towers with the aim of picking off defenders on the ramparts or causing random damage inside the walls.

To make a breach that could be stormed, legionaries had to get right up to the ramparts. This was the main use of the famous *testudo*.



CAVALRY HELMET

ROMAN AUXILIARIES AND CAVALRY

The auxiliaries were troops recruited from "barbarian" peoples, usually living within the borders of the Roman Empire but not enjoying the privilege of Roman citizenship. They provided extra manpower and specialist skills that the Roman army very much needed. In particular, they constituted the majority of the army's skirmishing light infantry and of its cavalry.

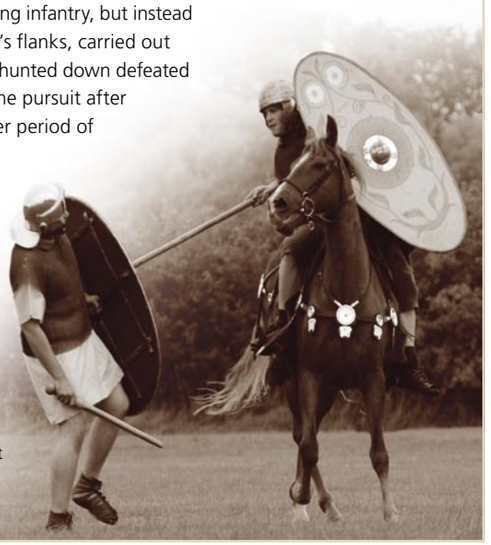
Like the legionaries, auxiliaries were expected to serve for 25 years. On completing the full term of service, their reward was the grant of Roman citizenship to them and their descendants. An auxiliary cohort was composed of recruits from a single area or ethnic group, but it was usually stationed away from its home area. Auxiliaries were paid less than legionaries and seem often to have been thrown into the most dangerous fighting as relatively dispensable troops. On the other hand, they were subject to less strict discipline, did less work – they were not involved in engineering or building – and carried less weight on the march.



AUXILIARY SHIELD

Many of the auxiliaries were trained to fight in a similar manner to the legionary, but some regions supplied specialist troops – the Balearics produced stone-slingers, for instance, and Syria was a source of bowmen. Most important were the expert horsemen from areas such as Batavia, Pannonia, and Thrace. They were the cream of the Roman cavalry, armed with spears, javelins, and the *spatha* – a longer sword than the infantry gladius. Although they did not have stirrups, their horned saddles served to hold them firmly on their mounts in combat.

Roman cavalry did not usually act as shock troops charging infantry, but instead defended the army's flanks, carried out encirclements, and hunted down defeated enemy soldiers in the pursuit after a victory. In the later period of the Empire, cavalry probably became more numerous and played an increasingly important role in battle.



Cavalry training
Cavalry training in the use of the spear is demonstrated by re-enactors. Note that Roman horsemen did not have stirrups.